Time change rigidity for unipotent flows

Joint work with Elon Lindenstrauss

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Isomorphism

Basic setting

$(X, \mathscr{B}, \mu)$, $(Y, \mathscr{C}, \nu)$ —standard probability Borel spaces; $T_t : X \to X$, $S_t : Y \to Y$ —measure-preserving ergodic flows.

Isomorphism relation

T_t and S_t are isomorphic through ψ, defined as $T_t \sim S_t$, if for $t \in \mathbb{R}$: $\psi \circ T_t = S_t \circ \psi$, where $\psi : X \to Y$ is invertible (mod 0) and $\psi_* \mu = \nu$.

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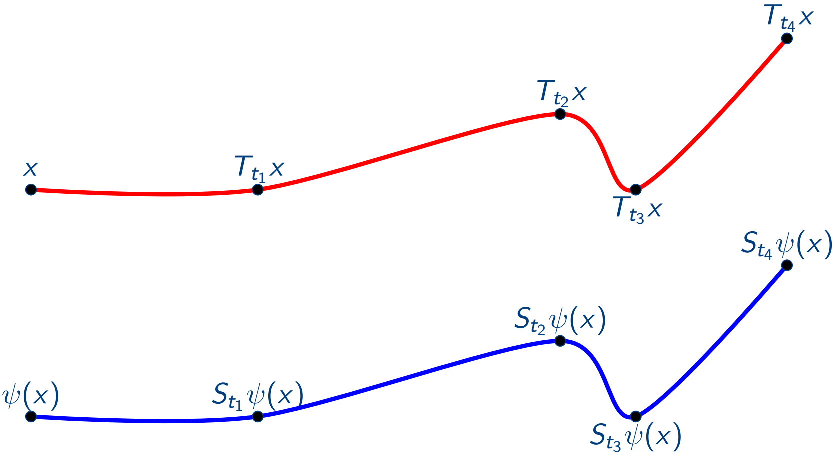

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Example

Entropy can be used to distinguish Bernoulli shifts among isomorphism equivalence class: $(\frac{1}{2}, \frac{1}{2})$ and $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$. The entropy of $(\frac{1}{2}, \frac{1}{2})$ is $\log 2$ and the entropy of $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ is $\log 3$.

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Isomorphic rigidity for unipotent flows

For $i = 1, 2$:

- G_i is a connected semisimple Lie group, $\mathfrak{g}_i$ is the corresponding Lie algebra and $\Gamma_i \subset G_i$ is a lattice in such that G_i acts faithfully on G_i/Γ_i;
- $U_i \in \mathfrak{g}_i$ is such that ad_{U_i} is nilpotent, where $\mathrm{ad}_{U_i} : \mathfrak{g}_i \to \mathfrak{g}_i$ and $\mathrm{ad}_{U_i}(V_i) = [U_i, V_i]$;
- $u_t^{(i)}$ acts ergodically on $(G_i/\Gamma_i, m_i)$, where $u_t^{(i)}(g\Gamma_i) = \exp(tU_i)g\Gamma_i$ and m_i is the probability measure defined on G_i/Γ_i induced from the Haar measure on G_i.

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Isomorphic rigidity for unipotent flows (Ratner, Witte)

If $u_t^{(1)}$ is isomorphic to $u_t^{(2)}$ through ψ, then there exist $c \in G_2$ and an isomorphism $\phi : G_1 \to G_2$ such that $\phi(\Gamma_1) = c^{-1}\Gamma_2 c$ and $\psi(g\Gamma_1) = \phi(g)c\Gamma_2$ for m_1-a.e. $g\Gamma_1 \in G_1/\Gamma_1$.

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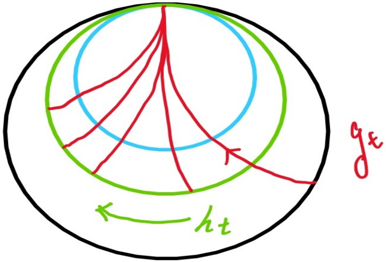

Related results

- Rigidity of horocycle flows&products of horocycle flows, M. Ratner, 1982&83;
- Rigidity of zero entropy affine maps and translations, D. Witte, 1985&87;
- Horocycle flows on the variable negative curvature space, J. Feldman and D. Ornstein, 1987;
- Horospherical foliations of two compact manifolds of constant negative curvature, L. Flaminio, 1987;
- Horospherical foliations on some bundles other than unit tangent bundle, D. Witte, 1989;
- Geometrically finite groups, Flaminio and Spatzier, 1990;
- Rigidity of unipotent flows as a corollary of measure rigidity, M. Ratner, 1990.

Kakutani equivalence

Kakutani equivalence, S. Kakutani 1943

T_t and S_t are Kakutani equivalent, defined as $T_t \stackrel{K}{\sim} S_t$, if they are measurably isomorphic to special flows over the same automorphism: there exist automorphism U and roof functions $f_1, f_2 \in L^1_+$ such that $T_t \sim U_t^{f_1}$ and $S_t \sim U_t^{f_2}$.

Time-change

For S_t and $\alpha \in L^1_+(Y,\nu)$, the time change of S_t is defined as $S_t^\alpha(y) = S_{\tau(y,t)}(y)$ for $y \in Y$, where $\int_0^t \alpha(S_\ell y) d\ell = \tau(y,t)$.

An equivalent definition: time-change

$T_t \stackrel{K}{\sim} S_t$ through $\psi : X \to Y$ if there exists $\alpha \in L^1_+(Y,\nu)$ such that $T_t \sim S_t^\alpha$ through ψ. We say τ is the time change cocycle corresponding to the Kakutani equivalence ψ.

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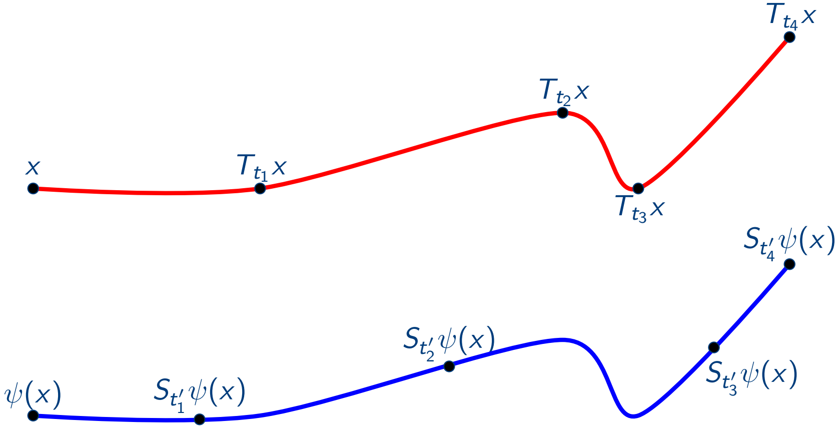

Why Kakutani equivalence?

Preserves the order on orbits

If $T_t \overset{K}{\sim} S_t$ through ψ, then for $x_1, x_2 \in X$, if $x_1 = T_{t_1} x_2$ for some $t_1 > 0$, we have $\psi(x_1) = S_{t_2}\psi(x_2)$ for some $t_2 \geqslant 0$.

Entropy class preserving:

Kakutani equivalence preserves zero entropy, positive entropy and infinite entropy classes due to Abramov formula.

Uncountably many non equivalent classes:

Uncountably many pairwise non-Kakutani equivalent systems, by D. S. Ornstein, D. Rudolph and B. Weiss, “Equivalence of measure preserving transformations”, 1976.

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Loosely Kronecker systems

Majorization

T_t majorizes S_t if there exists an ergodic measure preserving flow $U_t \overset{K}{\sim} T_t$ such that S_t is Kakutani equivalent to a measurable factor of U_t.

Loosely Kronecker systems (zero entropy loosely Bernoulli or standard)

Define $R_t^\alpha(x,y) = (x+t, y+t\alpha)$ for $\alpha \in \mathbb{R} \setminus \mathbb{Q}$, S_t is a loosely Kronecker flow if $S_t \overset{K}{\sim} R_t^\alpha$.

Characterization of loosely Kronecker systems

S_t is a loosely Kronecker flow iff any ergodic measure preserving flow majorizes S_t.

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Loosely Kronecker systems

- closed under factors, J. Feldman, 1976, A. Katok, 1976;
- closed under inverse limits and compact extensions — all nil-systems, A. Katok, 1976; D. S. Ornstein, D. Rudolph and B. Weiss, 1976;
- finite rank systems — all translation flows, D. S. Ornstein, D. Rudolph and B. Weiss, 1976;
- horocycle flow, M. Ratner, 1978;
- some Gaussian systems, T. De La Rue, 1996;
- local rank one system, S. Ferenczi, 1997.

Non-loosely Kronecker examples

- non-loosely Kronecker example, J. Feldman, 1976; D. S. Ornstein, D. Rudolph and B. Weiss, 1976;
- smooth non-loosely Kronecker example, A. Katok, 1977;
- product of horocycle flows, M. Ratner, 1979;
- a zero entropy Veršik process that is not loosely Kronecker, A. Rothstein, 1980;
- some Gaussian systems, T. De La Rue, 1996;
- product of Kochergin flows, A. Kanigowski, W., 2017;
- product of two staircase rank one transformations, A. Kanigowski, T. De La Rue, 2018.

Time changes rigidity for unipotent flows

- Horocycle flows are loosely Kronecker, M. Ratner, 1978;
- Product of horocycle flows are not loosely Kronecker, M. Ratner, 1979;
- Rigidity of time change of horocycle flow with some regularity conditions on time changes, M. Ratner, 1986, 1987;
- Rigidity of time changes of unipotent flows on $\mathrm{SO}(n, 1)$ with some regularity conditions on time change changes, S. Tang, 2020.
- Rigidity of smooth time changes of unipotent flows on a connected semisimple linear group with a graph-joining assumption (can be dropped in the case $u_t = h_t \times \mathrm{id}$), Artigiani–Flaminio–Ravotti, 2022.

Open problems

- For $i = 1, 2$, let $\Gamma_i \subset \mathrm{SL}_2(\mathbb{R})$ be a lattice, ν_i the probability measure of $M_i = \mathrm{SL}_2(\mathbb{R})/\Gamma_i$ induced by Haar measure on $\mathrm{SL}_2(\mathbb{R})$ and $h_t^{(i)}$ the horocycle flow acting on (M_i, ν_i).

Ratner's problem, ICM, 1994, Zürich, Proceedings of the International Congress of Mathematicians, p.179, problem 1 and 2

- Is the flow $h_t^{(1)} \times h_t^{(1)}$ acting on $(M_1 \times M_1, \nu_1 \times \nu_1)$ Kakutani equivalent to $h_t^{(2)} \times h_t^{(2)}$ acting on $(M_2 \times M_2, \nu_2 \times \nu_2)$?
- Are these isomorphism rigidity properties holding for smoothly time changes of one-parameter unipotent subgroups' actions?

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Our setting

For $i = 1, 2$:

- G_i is the identity component of a real semisimple algebraic group without compact factors, $\mathfrak{g}_i$ is the corresponding Lie algebra and $\Gamma_i \subset G_i$ is a lattice such that G_i acts faithfully on G_i/Γ_i;
- $U_i \in \mathfrak{g}_i$ is such that ad_{U_i} is nilpotent, where $\mathrm{ad}_{U_i} : \mathfrak{g}_i \to \mathfrak{g}_i$ and $\mathrm{ad}_{U_i}(V_i) = [U_i, V_i]$;
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- By Jacobson-Morozov theorem, for any nontrivial ad-nilpotent element $U_2 \in \mathfrak{g}_2$, there is a $\mathfrak{sl}_2$-triple U_2, V_2, X_2 such that $[U_2, V_2] = X_2$, $[X_2, U_2] = 2U_2$ and $[X_2, V_2] = -2V_2$. Moreover, let $a_s^{(2)}(g\Gamma_2) = \exp(sX_2)g\Gamma_2$.

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Main theorem (E. Lindenstrauss and W.)

If $u_t^{(1)} \overset{K}{\sim} u_t^{(2)}$, then either:

- $\mathfrak{g}_i = \mathfrak{sl}_2(\mathbb{R}) \times \mathfrak{g}_i'$ and the generator of $u_t^{(i)}$ is in the form $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \times 0 \in \mathfrak{g}_i$ for $i = 1, 2$;
- There exist an isomorphism $\phi : G_1 \to G_2$ and $c \in G_2$ such that $\phi(\Gamma_1) = c\Gamma_2 c^{-1}$. Moreover, there exists $s_0 \in \mathbb{R}$ such that $a_{s_0}^{(2)} \circ \psi$ is cohomologous to an actual isomorphism, i.e.

$$a_{s_0}^{(2)} \psi(g\Gamma_1) = u_{w(g\Gamma_1)}^{(2)} \phi(g) c\Gamma_2,$$

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Our theorem does not have any regularity restrictions for the time changes and thus our theorem provides more rigidity than M. Ratner's original equstion.

Corollary (Answer for Problem 1)

If $G_i = \mathrm{SL}_2(\mathbb{R}) \times \mathrm{SL}_2(\mathbb{R})$ and $u_t^{(i)} = \left(\begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix} \right)$ for $i = 1, 2$, then $u_t^{(1)} \overset{K}{\sim} u_t^{(2)}$ if and only if there is a $c \in \mathrm{SL}_2(\mathbb{R}) \times \mathrm{SL}_2(\mathbb{R})$ such that $\Gamma_1 = c\Gamma_2 c^{-1}$.

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Corollary

- By Kanigowski–Vinhage–W. 18, we know that an ergodic unipotent flow u_t of G/Γ is loosely Kronecker if and only if $\mathfrak{g} = \mathfrak{sl}_2(\mathbb{R}) \oplus \mathfrak{g}'$ and u_t is generated by $U = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \times 0.$

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In 1980, M. Ratner created an invariant $e(T_t, u)$ to evaluate the complexity among Kakutani equivalence class quantitatively.

$\bar{f}$-metric

- $\mathcal{P}$: a finite measurable partition of X;
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- $I_R(x) = \{T_s x : s \in [0, R]\}$.

$I_R(x)$ and $I_R(y)$ are called $(\varepsilon, \mathcal{P})-$matchable if there exist subsets $A = A(x, y), A' = A'(x, y) \subset [0, R]$, $l(A), l(A') > (1 - \varepsilon)R$ and an increasing measure preserving map $h = h(x, y)$ from A onto A' such that $\mathcal{P}(T_t x) = \mathcal{P}(T_{h(t)} y)$, $\forall t \in A$. Then define $\bar{f}-$metric as

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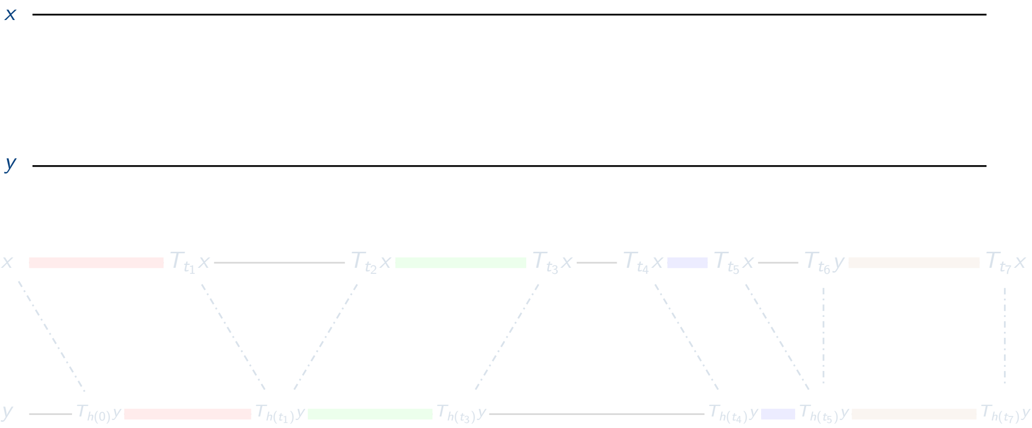

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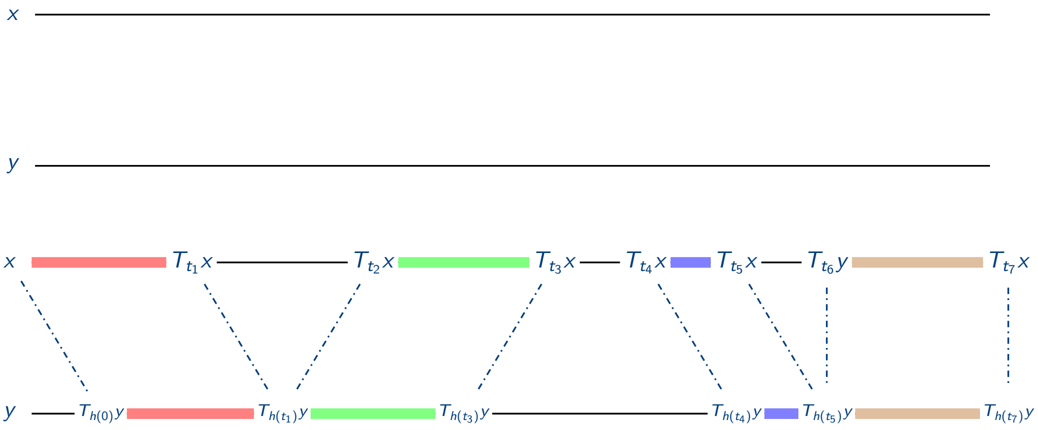

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- $\mathscr{F}$: the family of all nondecreasing function $u : \mathbb{R}^+ \to \mathbb{R}^+$ such that $\lim_{t\to+\infty} u(t) = +\infty$;
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There exist a full measure subset $K_1 \subset G_1/\Gamma_1$ such that for every $x \in K_1$, there exists $t \in \mathbb{R}$ such that

$$\tilde{\psi}(a_1^{(1)}x) = a_1^{(2)} u_t^{(2)}(\tilde{\psi}(x)),$$

where $\tilde{\psi} = c\psi$ and $c \in C_G(u_t^{(2)}) \setminus \{u_t^{(2)}\}$.

Key ideas of the proof

- Let $\tilde{\psi}_n = a_{-n}^{(2)}\tilde{\psi}(a_n^{(1)}x)$, we want to show that along a subsequence $\{n_i\}_{i\in\mathbb{N}}$ (may depends on x) with full density, $\{\tilde{\psi}_{n_i}(x)\}$ convergent for m_1-a.e. $x \in G_1/\Gamma_1$;
- Then we show the limiting abstract map $\bar{\psi}$ of $\tilde{\psi}_{n_i}(x)$ is a measurable isomorphism between $u_t^{(1)}$ and $u_t^{(2)}$;
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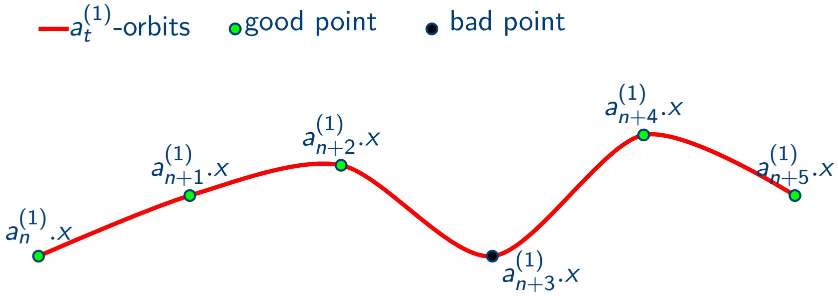
Key ideas of the proof

- Let $\tilde{\psi}_n = a^{(2)}_{-n}\tilde{\psi}(a^{(1)}_n x)$, we want to show that along a subsequence $\{n_i\}_{i\in\mathbb{N}}$ (may depends on x) with full density, $\{\tilde{\psi}_{n_i}(x)\}$ convergent for m_1-a.e. $x \in G_1/\Gamma_1$;
- Then we show the limiting abstract map $\bar{\psi}$ of $\tilde{\psi}_{n_i}(x)$ is a measurable isomorphism between $u^{(1)}_t$ and $u^{(2)}_t$;
- Finally we activate the isomorphism rigidity theorem for unipotent flows;

Key ideas of the proof



- In order to deal with "bad" iterations, we make small perturbations in B_ε^H.

L^2-ergodic theorem

L^2-ergodic theorem (E. Lindenstrauss and W.)

Let ρ be a unitary representation of $\mathbf{H} = \mathrm{SL}_2(\mathbb{R})$ on a Hilbert space $\mathcal{H}_\rho$ with no fixed vectors and a spectral gap. Then there exists $\varepsilon_1 > 0$ such that for any $\varepsilon \in (0, \varepsilon_1)$ there is a $C(\varepsilon)$ so that

$$\sum_{n=1}^{+\infty} \left\| \frac{1}{m_{\mathbf{H}}(B_\varepsilon^{\mathbf{H}})} \int_{B_\varepsilon^{\mathbf{H}}} \rho(ha_{-n})v \, dm_{\mathbf{H}}(h) \right\|^2 < C(\varepsilon)\|v\|^2$$

for any $v \in \mathcal{H}_\rho$, where $m_{\mathbf{H}}$ is the Haar measure on $\mathbf{H}$ and $B_\varepsilon^{\mathbf{H}}$ is the ε-ball in $\mathbf{H}$ around identity.

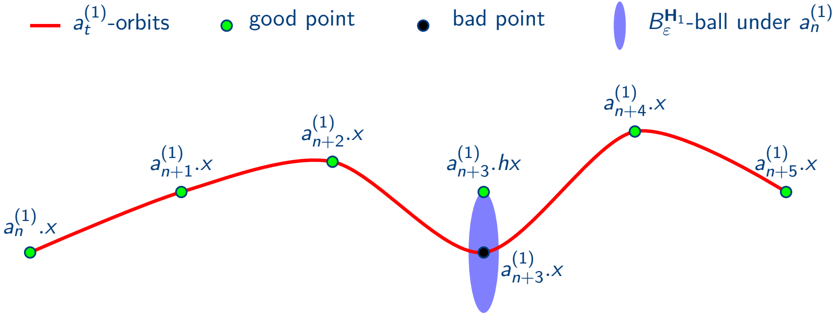
Corollary

Corollary (E. Lindenstrauss and W.)

Let G be the identity component of a real semisimple algebraic group without compact factors, $\Gamma < G$ a lattice with m the probability measure on G/Γ induced by Haar measure on G. Let $H < G$ be locally isomorphic to SL_2. Given ε sufficiently small, then for any $f \in L^2(G/\Gamma, m)$, m-a.e. $x \in G/\Gamma$, we have

$$\lim_{n \to +\infty} \frac{1}{m_H(B_\varepsilon^H)} \int_{B_\varepsilon^H} f(a_n h.x) dm_H(h) = \int_{G/\Gamma} f dm.$$

Key ideas of the proof



Joyeuse Retraite !

Congratulations to Professor Flaminio and best wishes for the next chapter.

Thank you!

Questions & discussion